

JEE MAIN REVISION MASTER: TRIGONOMETRIC RATIOS & IDENTITIES 2026 2 QUESTIONS 3.33% WEIGHTAGE

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 13 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	Kcet Trigonometric Ratios And Identities - Core Application	13 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Kcet Trigonometric Ratios And Identities - Core Application (High Yield — 13 PYQs)

Core Principle: Key problem pattern in Kcet Trigonometric Ratios And Identities.

■ ■ Caution / JEE Pitfall: Watch for algebraic and sign errors.

■ Speed Trick: Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "If α and β are acute angles such that $\alpha - \beta$ and $\alpha + \beta$ satisfy the equation $\tan^2\theta - 4\tan\theta + 1 = 0$, t..." [Ans: A]
- [2026 Shift] Q#2 (MCQ): "The maximum value of $\sin(x + (\pi/6)) + \cos(x + (\pi/6))$ is attained at $x = \dots$ " [Ans: D]
- [2025 Shift] Q#3 (MCQ): "Which of the following is not correct?..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

If α and β are acute angles such that $\alpha - \beta$ and $\alpha + \beta$ satisfy the equation $\tan^2\theta - 4\tan\theta + 1 = 0$, then α and β are respectively,

- (A) $45^\circ, 30^\circ$ **[Correct]**
- (B) $30^\circ, 45^\circ$
- (C) $30^\circ, 60^\circ$
- (D) $60^\circ, 45^\circ$

Correct Answer: A

Solution Analysis: Put $x = \tan\theta$. The given equation is $\tan^2\theta - 4\tan\theta + 1 = 0$ So, $x^2 - 4x + 1 = 0$ Using the quadratic formula, $x = \frac{4 \pm \sqrt{16-4}}{2} = \frac{4 \pm 2\sqrt{3}}{2} = 2 \pm \sqrt{3}$ Now, $\tan 15^\circ = 2 - \sqrt{3}$ and $\tan 75^\circ = 2 + \sqrt{3}$ So the two angles satisfying the equation are 15° and 75° . Here the angles are $\alpha - \beta$ and $\alpha + \beta$. Since $\alpha + \beta$ is greater than $\alpha - \beta$, $\alpha - \beta = 15^\circ$ and $\alpha + \beta = 75^\circ$. Adding both equations, $2\alpha = 90^\circ$ $\alpha = 45^\circ$ Now, $\alpha - \beta = 15^\circ$